

Progress Report: Automated Fake News Detection — Bangla BiLSTM, Multimodal, and TruthLens Integration

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Abstract—This report consolidates progress across three related fake news detection efforts: (1) a Bangla-language BiLSTM text classifier ported from Google Colab to a local Windows/VS Code environment, (2) integration of an existing multimodal (text+image) fake news detector into the TruthLens system, and (3) hands-on Colab experimentation with the same multimodal model at increasing dataset scale. Together these efforts cover environment setup, dataset engineering, source-code adaptation, and initial training/evaluation results, and identify the key blockers that define the next phase of work.

I. OBJECTIVE

The overall goal is to build a working fake news detection pipeline for TruthLens using two complementary models: a lightweight Bangla-text BiLSTM classifier runnable on commodity Windows hardware, and a multimodal (text, image, metadata) classifier adapted from an existing open-source repository. Work spans porting Colab notebooks to local execution, resolving dependency and dataset incompatibilities, and producing initial trained checkpoints for evaluation.

II. SYSTEM CONFIGURATIONS

A. Bangla BiLSTM (Local)

Intel Evo Core i5, Intel Iris (shared 4GB) GPU, 8GB RAM, Windows 11; Python 3.10.11 in a venv; TensorFlow (CPU); VS Code with the Jupyter extension.

B. Multimodal Model (Colab / TruthLens)

Python 3.10.11 (required for dependency compatibility) in a venv; VS Code and Git for local integration work; Google Colab with GPU runtime (CUDA-enabled) used for actual training experiments; PyTorch, torchvision, Transformers, Sentence-Transformers, PyTorch-Lightning, pandas, Pillow.

III. WORK COMPLETED

A. Bangla BiLSTM Porting (Colab → Windows)

The Bangla-Fake-News-Detection repository was opened in VS Code and a virtual environment created and activated. The notebook kernel was switched from the global interpreter to the venv. Colab-specific paths (`/content/drive/MyDrive/...`) were rewritten as local relative paths, after which `data_file1/2.joblib.bz2` loaded successfully via `joblib.load()`, yielding `X_train`, `Y_train`, `X_test`, `Y_test`, `max_features`, and `bias`. The BiLSTM model (embedding, bidirectional LSTM, dense layers) was constructed and compiled without errors, with `batch_size=128`, `epochs=10`, class weights, and a `ModelCheckpoint` callback configured.

B. Multimodal Model Integration into TruthLens

The multimodal repository was cloned and its `requirements.txt` installed into a fresh venv; missing packages (`torch`, `torchvision`, `transformers`, `sentence-transformers`, `pandas`, `pillow`, `tqdm`) were identified and installed iteratively. Since the full Fakeddit dataset (text, images, metadata, and comment dialogue, hundreds of GB) was impractical for demonstration, a small custom dataset (three sample images plus train/valid/test TSVs with `id` and `clean_title / 2_way_label` columns) was built instead. Source code was patched to match: numeric IDs were converted to four-digit image filenames via `str(id).zfill(4)`, and preprocessing logic was modified to ignore Fakeddit-only fields (`created_utc`, `domain`, `hasImage`, `image_url`) absent from the custom set.

C. Multimodal Training Experiments (Colab, GPU)

In parallel, the original Multimodal-Fake-News-Detection repository was patched for modern library versions: `yaml.load` calls were made safe via `yaml.safe_load`; `dataloader.py` was updated to tolerate truncated images and to record dataset modality; and the PyTorch-Lightning Trainer calls in `run_training.py` / `run_evaluation.py` were migrated from the deprecated `gpus/strategy="dp"` API to `accelerator="gpu"`,

`devices=1`. The removed `test_epoch_end` hook was reimplemented as `on_test_epoch_end` with manually accumulated outputs. Training was then run at increasing scale, summarized in Table 1.

Table 1: Multimodal Model: Scale vs. Test Accuracy

Train / Test size	Epochs	Batch	Test Acc.
150 / 50	1	8	0.5625
1200 / 300	3	8	0.6125
4000 / 1000	5	16	0.7020

Test accuracy improved from 56.3% at the smallest scale to 70.2% once the training set was expanded to 4,000 samples over 5 epochs, indicating the pipeline is functioning correctly and is data-limited rather than broken.

IV. PROBLEMS IDENTIFIED

BiLSTM (Windows): training via `model.fit(...)` printed Epoch 1/10 but never advanced past batch 0 for 10–12 minutes; the Python process remained alive (no exception), and the Jupyter kernel eventually became unresponsive to interrupt requests. Suspected causes include residual Colab-specific code, lingering Google Drive checkpoint paths, CPU-only TensorFlow execution on limited hardware, and large in-memory dataset loading on an 8GB-RAM machine.

TruthLens Multimodal Integration: beyond the expected issues of missing Fakeddit-only metadata and image-filename mismatches, edits to `dataloader.py` introduced mixed tabs/spaces, producing an `IndentationError: expected an indented block after function definition` that currently blocks further execution. Several deprecated APIs and version-related incompatibilities (consistent with the age of the original repository) were also encountered, mirroring issues later resolved in the Colab experiments (Section III-C).

V. CURRENT STATUS

Table 2: Status Summary Across Both Models

Component	Status
BiLSTM: env./VS Code/dataset/model build	Completed
BiLSTM: training	Stuck (Epoch 1)
BiLSTM: evaluation / inference	Pending
Multimodal: env., deps., demo dataset	Completed
Multimodal: dataloader.py (TruthLens copy)	Blocked (indentation)
Multimodal: Colab training/evaluation	Completed (70.2% acc.)
TruthLens integration of trained model	Pending

VI. PLANNED WORK

BiLSTM: audit remaining Colab-specific code and Drive paths, verify CPU execution and callback configuration, and reduce memory footprint before re-attempting training; then evaluate (accuracy, precision, recall, F1, confusion matrix) and run inference on new articles. **Multimodal/TruthLens:** repair `dataloader.py` indentation and port the fixes already validated in the Colab experiments, complete pre-processing for the custom dataset, and re-run `run_training.py` / `run_evaluation.py` locally. Subsequently scale to a larger (or full) Fakeddit sample, and integrate the trained checkpoint into the TruthLens VLA + AR application. Optional stretch goals include modularizing notebooks into scripts, adding a `requirements.txt`, and building a lightweight Streamlit/Flask demo interface.

VII. CONCLUSION

Both detection pipelines have cleared environment setup and initial model construction. The multimodal model has already produced a working, improving classifier (up to 70.2% test accuracy) on Colab, giving a validated reference implementation to port back into the TruthLens codebase. The Bangla BiLSTM model builds correctly but is blocked by an unresolved training hang on local Windows hardware, and the TruthLens copy of the multimodal dataloader is blocked by a code-formatting error — both are the immediate priorities for the next work phase.